



SENTINEL AI - Quick Start Guide

Get SENTINEL running in 10 minutes!

Prerequisites

- Python 3.10+
 - Anthropic API Key ([Get one here](#))
 - 8GB RAM minimum
 - Docker (optional but recommended)
-

Option 1: Quick Start (No Docker)

Step 1: Clone & Setup

```
bash

git clone https://github.com/yourusername/sentinel-ai.git
cd sentinel-ai

# Run automated setup
chmod +x setup.sh
./setup.sh

# Or manually:
python3 -m venv venv
source venv/bin/activate
pip install -r requirements.txt
```

Step 2: Configure

```
bash
```

```
# Copy environment template
```

```
cp .env.example .env
```

```
# Edit with your API key
```

```
nano .env
```

Minimum required configuration:

```
bash
```

```
ANTHROPIC_API_KEY=sk-ant-your-key-here
```

```
POSTGRES_PASSWORD=your_password
```

```
NEO4J_PASSWORD=your_password
```

Step 3: Start Services (Simple Mode)

```
bash
```

```
# Terminal 1: Start API
```

```
python -m uvicorn api.main:app --host 0.0.0.0 --port 8000
```

```
# Terminal 2: Start Dashboard
```

```
streamlit run dashboard/app.py
```

```
# Terminal 3 (Optional): Run main orchestrator
```

```
python main.py
```

Access:

- Dashboard: <http://localhost:8501>
 - API Docs: <http://localhost:8000/docs>
-

Option 2: Docker (Recommended)

One Command Setup

```
bash

# Clone and start
git clone https://github.com/yourusername/sentinel-ai.git
cd sentinel-ai
cp .env.example .env

# Edit .env with your API key
nano .env

# Start everything
docker-compose up -d

# Check logs
docker-compose logs -f
```

Access:

- Dashboard: <http://localhost:8501>
- API: <http://localhost:8000>
- Neo4j Browser: <http://localhost:7474>

Stop Services

```
bash

docker-compose down
```

First Run Checklist

- ☒ **Verify Installation**

```
bash
```

```
# Check API health
```

```
curl http://localhost:8000/health
```

```
# Check dashboard
```

```
open http://localhost:8501
```

✓ Test Detection

The system will automatically start monitoring. To test:

1. **View Dashboard:** Open <http://localhost:8501>
2. **Check Agent Status:** All 5 agents should show status
3. **Monitor Incidents:** Watch the "Live Monitoring" tab

✓ Simulate Attack (Optional)

```
bash
```

In Python console

```
from agents.watcher import WatcherAgent
from core.models import ThreatDetection, AttackType, SeverityLevel
import asyncio
```

```
async def test():
    threat = ThreatDetection(
        source_ip="192.168.1.100",
        destination_ip="10.0.0.1",
        attack_type=AttackType.SQL_INJECTION,
        severity=SeverityLevel.HIGH,
        confidence_score=0.95,
        raw_data={"payload": "admin' OR '1'='1'--"},
        indicators=["sql_injection"]
    )
    print("Threat created for testing")

asyncio.run(test())
```

Common Issues & Solutions

Issue: "Module not found"

bash

Ensure virtual environment is activated

source venv/bin/activate

Reinstall dependencies

pip **install** -r requirements.txt

Issue: "Database connection failed"

bash

Check if PostgreSQL is running

`docker-compose ps`

Or start manually

`docker-compose up -d postgres`

Verify credentials in .env match docker-compose.yml

Issue: "API key invalid"

bash

Verify your Anthropic API key

`echo $ANTHROPIC_API_KEY`

Or check .env file

`cat .env | grep ANTHROPIC`

Issue: "Port already in use"

bash

Change ports in .env or docker-compose.yml

Default ports: 8000 (API), 8501 (Dashboard), 5432 (Postgres)

Find what's using port

`lsof -i :8000`

Understanding SENTINEL

The 5 AI Agents

1.  **Watcher:** Detects threats in real-time
2.  **Investigator:** Analyzes attacks

3. 💰 **Healer:** Remediates automatically
4. 💰 **Economist:** Calculates financial impact
5. 📖 **Librarian:** Documents & learns

Key Features

- **Autonomous Response:** No human intervention needed
 - **Real-time Analysis:** Threats analyzed in seconds
 - **Financial Impact:** Know the cost of every breach
 - **Self-Learning:** Gets smarter with every incident
 - **Compliance Ready:** Auto-generates SOC2, ISO 27001 reports
-

Next Steps

1. Configure Monitoring

Edit `core/config.py` to customize:

- Honeypot ports
- Auto-remediation settings
- Alert thresholds
- Compliance frameworks

2. Explore the Dashboard

- **Live Monitoring:** Real-time threat feed
- **Analytics:** Attack trends and patterns
- **Incidents:** Detailed investigation logs
- **Agents:** System health monitoring

3. API Integration

```
python

import requests

# Get incidents
response = requests.get("http://localhost:8000/api/incidents")
incidents = response.json()

# Get dashboard stats
response = requests.get("http://localhost:8000/api/dashboard")
stats = response.json()
```

4. Enable Secret Weapons

Uncomment in `core/config.py`:

```
python

# Dark Web Monitoring
DARK_WEB_ENABLED = True

# Predictive Threat Intelligence
PREDICTIVE_ANALYSIS = True
```

Production Deployment

Security Hardening

1. **Change default passwords** in `.env`
2. **Enable HTTPS** with reverse proxy (nginx/traefik)
3. **Set up firewall rules**
4. **Configure log rotation**

5. Enable authentication on dashboard

Scaling

```
yaml

# docker-compose.yml

services:
  sentinel-api:
    deploy:
      replicas: 3
    environment:
      - WORKERS=4
```

Monitoring

- Enable Prometheus metrics: `METRICS_ENABLED=true`
 - Set up Grafana dashboards
 - Configure alerting (email, Slack, PagerDuty)
-

Performance Tuning

For High Traffic

```
bash

# In .env

MAX_AGENT_ITERATIONS=5
AGENT_TIMEOUT=120
HONEYPOT_ENABLED=false # Disable if not needed
```

For Low Resources

```
bash
```

```
# In .env
```

```
AUTO_REMEDIATION=false
```

```
DARK_WEB_ENABLED=false
```

```
LOG_LEVEL=WARNING
```

Getting Help

Documentation

- [Architecture Guide](#)
- [API Reference](#)
- [Agent Development](#)

Community

- GitHub Issues: [Report bugs](#)
- Discord: [Join community](#).
- Email: support@sentinelai.dev

Troubleshooting

```
bash
```

```
# Check logs
```

```
tail -f logs/sentinel_*.log
```

```
# Test database connection
```

```
python -c "from core.database import db_manager; print('OK')"
```

```
# Verify API
```

```
curl http://localhost:8000/health
```

Demo Mode

Want to test SENTINEL with simulated attacks?

```
bash
```

```
# Run demo mode
```

```
python demo.py
```

```
# This will:
```

```
# 1. Generate sample attacks
```

```
# 2. Show real-time detection
```

```
# 3. Demonstrate auto-remediation
```

```
# 4. Create sample reports
```

What's Next?

For the Lyzr Competition:

1.  System running
2.  Dashboard accessible
3.  Incidents being detected
4.  Reports generated
5.  Demo video recorded

Create Your Demo:

```
bash
```

Record dashboard showing:
1. Live threat detection (30 sec)
2. Agent coordination (30 sec)
3. Financial impact calculation (30 sec)
4. Auto-remediation (30 sec)
5. Knowledge base growth (30 sec)

Total: 2.5 minutes of pure innovation

Success Metrics

After 24 hours of operation, you should see:

-  10+ incidents detected
-  5+ auto-remediated
-  3+ playbooks created
-  Financial impact calculated
-  Compliance reports generated

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 **You're All Set!**

SENTINEL is now protecting your infrastructure autonomously.

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